Oral Exam Preparation

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1 Syllabus

- Foundations of Algebraic Geometry
  - Schemes (Hartshorne II.1-II.8)
  - Cohomology (Hartshorne III)
  - Curves (Hartshorne IV)
  - Complex Geometry (Huybrechts 1-4)
  - Syzygies (Ein-Lazarsfeld *Lectures on the Syzygies and Geometry of Algebraic Varieties* 1-6)
- Derived/Triangulated Categories (in algebraic geometry or otherwise)
  - Definitions (Gelfand Manin *Methods in Homological Algebra* II-IV)
  - Exceptional Collections/Tilting Bundles (Craw's *Explicit Methods for Derived Categories of Sheaves* Lectures 1-2,)
  - In Algebraic Geometry (Huybrechts *FM Transforms* 1-8)
- Algebra
  - Representation Theory (Fulton and Harris Part III)
  - Quivers (Assem Skowronski Simson Chapters I-II and some of III)

2 Preparation

Relevant notes:

- for schemes: divisors, projective_space, properties_of_morphisms
- for curves: curves
- for syzygies: syzygies

- for derived categories: stability_conditions, beilinson, tilting, hpd
- for representation theory: representation_theory

TODO:

- write notes about canonical bundle SESs (a big weak point for me)
- write notes about proving Serre duality/cohomology base change
- make thorough curve notes
- write notes for semiorthogonal collection
- organize notes around Fourier Mukai Transforms
- ideally prove